

Suvanth Erranki

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EDUCATION

Carnegie Mellon University, School of Computer Science – GPA: 4.0/4.0 Expected May 2029
Bachelor of Science, Computer Science Pittsburgh, PA
Relevant Coursework: Matrices and Linear Transformations, Mathematical Foundations of CS, Principles of Imperative Computation, Real Analysis (at Princeton), Number Theory (at Princeton)

ACTIVITIES & EXPERIENCE

Software Engineer Intern (College Tech I) Jun. 2026 – Aug. 2026
Lockheed Martin Moorestown, NJ

- Built a cost-analysis program allowing Finance, Program Managers, and Data Analysts to query an aerospace parts dataset in natural language, translating prompts into DuckDB SQL via an LLM wrapper and executing queries in pandas to return a filtered subset.
- Modeled the filtered query data as a cost-flow subgraph and applied Kahn’s topological sort algorithm to trace dependencies and sum cumulative cost/labor deltas.
- Rendered results as interactive cost-flow graphs with per-category breakdowns in a Plotly/Dash dashboard with export support. Displayed problem nodes with >5% weekly EAC increase for review.

Lead Organizer, West Windsor-Plainsboro Programming Competitions Feb. 2024 – Jun. 2026

- 501(c)(3) sponsored by Jane Street and HRT; hosted 3 programming competitions for 500+ students (latest at TCNJ); distributed \$10K+ in cash prizes, raffles, and merchandise. wwppc.org/wwpit

AWARDS

USA Computing Olympiad Platinum Division & 2026 Proctored Contest Invitee (1 of 75)	Mar. 2026
MIT Battlecode 2025 Finalist – 1st/113 HS Teams, 7th/504 Overall (Livestream Postmortem)	Jan. 2025
1st Place — Cornell HS Programming Contest (75 teams)	Mar. 2025
1st Place — UPenn PClassic Advanced (50 teams)	Apr. 2025
1st Place — Lockheed Martin Code Quest Advanced (30 teams)	Feb. 2026

PROJECTS

WWP Programming Competitions Judgehost | *TypeScript, Node.js, Linux*

- Sandboxed submissions with the [isolate](#) library, enforcing per-language time and memory limits; added custom testcase generators, custom checkers, interactive problems, and Polygon-format problem import.
- Took over the job queue code distributing submissions across eight grading servers after our lead developer graduated; improved error handling and fixed requeue and timeout bugs.
- Returned verdicts for 5K+ contest submissions in C/C++, Java, and Python/PyPy.

Chess Engine “Kinetic Hero” | *C++23* | [Source Code](#)

- Generated moves with magic bitboards and precomputed attack tables, reaching 60M nodes/sec on standard perft suites (M2 Pro, single core).
- Implemented Monte Carlo Tree Search with PUCT selection, storing the tree in a preallocated array to avoid allocation at runtime.

SKILLS

Languages: C++, Python, C, Java, JavaScript, TypeScript, HTML/CSS
Systems & Tools: Linux, Docker, Git, PostgreSQL, MySQL, Bash, Vim
ML/Data: PyTorch, NumPy, pandas, Matplotlib
Frameworks: React, Node.js, Vue